Economic Integration in Canada: The Impact of Immigration Duration on Wage Gaps Between Natives and Non-Natives

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April 29, 2025

Abstract

This paper explores wage disparities between immigrants and native-born Canadians, focusing on how wage gaps evolve with time spent in the country. Using data from the 2022 Canadian Income Survey, I estimate an OLS model with log-transformed hourly wages. Results show that immigrants earn about 19.4% less than natives, on average. However, the gap narrows as time since immigration increases, nearly disappearing after 30–39 years. The findings suggest that underpayment may be linked to unmeasured factors such as foreign credential recognition and language skills. These results highlight the importance of targeted policies to improve immigrant wage outcomes over time.

1 Introduction

Over just the past decade, Canada has welcomed over one million immigrants from a wide range of countries (Statistics Canada 2022), facilitated by the Immigration, Refugees and Citizenship Canada (IRCC) agency. Among the various immigration pathways, the skilled-worker program has emerged as the most significant, aiming to attract individuals with foreign work experience who are of working age. This policy responds to Canada's demographic challenge of an aging population by seeking to replenish the labour force with qualified workers. Most immigrants arrive as permanent residents and tend to settle in major urban centres such as Toronto, Vancouver, and Montreal (Statistics Canada 2021), where wage opportunities are relatively high. However, despite their qualifications and experience, many immigrants face challenges integrating into the Canadian labour market.

One of the most persistent issues is the immigrant wage gap, which is a form of hidden economic penalty that prevents newcomers from earning on par with their Canadian-born counterparts. This phenomenon is not unique to Canada, as similar wage disparities have been documented in other high-income countries such as the United States (Stewart and Dixon 2010) and Germany (Ingwersen and Thomsen 2021). Many reasons have been cited for this gap, but, most prominently, it is the difference in language proficiency and the lack of recognition for foreign credentials that are driving factors (The Conference Board of Canada 2016).

This paper investigates the extent of the immigrant wage gap in Canada and how it evolves as immigrants integrate into the workforce. The structure of the paper is as follows: Section 2 reviews existing literature on the wage gap and immigrant labour market outcomes. Section 3 describes the dataset used, drawn from Statistics Canada, while Section 4 outlines the empirical strategy, which employs an Ordinary Least Squares (OLS) model to estimate the wage gap. The results of the analysis are presented in Section 5, followed by a broader discussion of the implications for immigration policy and labour market equity in Section 6.

2 Literature Review

Lamb, Banerjee, and Verma (2021) present a framework that examines the rise of non-standard employment (such as part-time and temporary work) in Canada over the past few decades. The study explores the wage disparities between standard and non-standard employment, highlighting the prevalence of low wages within non-standard work, which is often viewed as a stepping stone to full-time, permanent positions. The research emphasizes that non-standard employment has a particularly significant impact on minority groups. Lamb's analysis draws on data from the Labour Force Survey spanning from 2006 to 2016 and incorporates variables related to human capital, personal characteristics, employment, and geographical factors. The findings reveal that immigrants face a 30% higher likelihood of being employed in non-standard jobs compared to their Canadian-born counterparts. Additionally, the study identifies a persistent 19% wage gap between immigrants and natives across both genders. Furthermore, Lamb's results suggest that immigrants experience a 0.5% increase in their hourly wage for each year spent in Canada.

Ingwersen and Thomsen (2021) develop a model that divides the population into three groups (natives, naturalized immigrants, and foreigners) to analyze wage gaps in Germany. The analysis draws on data from 1994 to 2015 and uses quantile regression along with a recentered influence function to estimate an endowment gap, which is a recalculated wage gap. The model incorporates various cultural and economic variables, with the study finding that these factors have the most significant impact on the wage gap. Additionally, the model considers the foreign qualifications of individuals. Ingwersen's findings reveal a wage gap of 14 to 18% between foreigners and natives, while the gap between naturalized immigrants and foreigners ranges from 10 to 16%, both in favor of natives. The wage gap is most significantly explained among higher wage deciles, although unexplained wage differences remain among lower deciles.

Ansah and Mueller (2021) examine the wage gap between immigrant and Canadian-born workers across both the private and public sectors, along with various subsectors, building upon prior literature in the field. The study discusses immigration as a strategy to address the challenges posed by Canada's aging population. Data for the analysis is sourced from the Canadian Labour Force Survey master files covering the period from 2006 to 2018. To refine the analysis, the study utilizes NAICS codes to categorize sub-sectors. The Blinder-Oaxaca decomposition method is employed to assess the wage gap. The results reveal that Canadian-born workers earn a 12% wage premium over their non-Canadian counterparts in the private sector, while the gap narrows to 10% in the public sector, with an average differential of 6% across federal, provincial, and municipal levels. Interestingly, 80% of the wage differential in the private sector remains unexplained, suggesting the possibility of discrimination or unrecognized disparities in the valuation of foreign credentials.

While these studies consistently identify an immigrant wage gap, few explore how the gap changes dynamically over time in a Canadian context using recent microdata. This paper fills that gap by adding a new perspective through well-known theories and methods.

3 Data

This paper analyzes data sourced from the Canadian Income Survey (Statistics Canada 2023a), which, in turn, sources some variables from the Labour Force Survey (Statistics Canada 2023b). Although the reference year for the data is 2022, the collection period spans from January 2023 to July 2023. This was done by Statistics Canada to align responses with the CRA tax year. This dataset includes a comprehensive range of information, covering individual, economic household, and census household-level data. In addition to income, tax, and labour data, the dataset also has demographic and geographic variables.

All data cleaning and transformation for this paper was carried out using the Tidyverse package (Wickham et al. 2019) in R (R Core Team 2024). The raw dataset contains over 96,000 observations and 210 variables. However, only the most relevant variables were selected for analysis (highest education level, immigrant status, sex, etc.). The focus was on individuals within the working-age population, specifically those aged between 15 and 64 inclusive. Paid employees with wages above zero were selected, excluding observations with invalid wage values. While there were many missing values and non-responses, some of the missing data were associated with valid skips, which occur when a question is not applicable to a respondent (e.g., employment status for students).

The variables in the data were transformed to enhance model clarity and consistency. The `province` variable was mapped from province codes to their corresponding abbreviations, and refactored to make Ontario the baseline province. Additionally, the variables `pop_level` and `age_level` were re-scaled to a 1–5 and 1–10 range, respectively, to standardize their measurement. The `female` and `full_time` variables were converted into binary indicators, where 1 represents the individual being female or employed full-time during the reference year. The `imm_length_level` variable was also adjusted so that natives were assigned a value of 0, while remaining the same for non-natives. Lastly, a new variable (`hourly_wage`) was created to better measure earnings by dividing total wages earned by the total number of hours worked in the reference year. Descriptions for all variables in the cleaned dataset can be found in the data dictionary in the appendix.

Table 1: Summary Statistics

Statistic	N	Mean	St. Dev.	Min	Median	Max
pop_level	30,171	2.97	1.28	0	3	4
age_level	30,171	4.82	2.62	0	5	9
female	30,171	0.50	0.50	0	0	1
highest_ed_level	30,171	2.03	0.92	0	2	3
full_time	30,171	0.86	0.35	0	1	1
hours	30,171	1,711.39	652.96	6	1,950	5,720
wages	30,171	65,698.48	50,549.66	125	57,500	625,000
imm	30,171	0.19	0.39	0	0	1
imm_length_level	30,171	0.43	1.05	0	0	5
hourly_wage	30,171	37.81	25.27	3.56	31.59	211.11

Table 1 indicates a fairly representative sample of the population. The average and median population levels suggest that respondents predominantly come from areas with populations ranging from 100,000 to 500,000 people, with 50% living in areas above 500,000. The median age is consistent with broader population estimates, falling within the 40–44 years range. Distribution of sex is balanced, with 50% women and 50% men, meaning there is proper representation. The majority of respondents have completed post-secondary education, with the average education level slightly above non-university certificates or diplomas. Summary statistics for the variable `full_time` shows that 86% of respondents have worked full-time at some point in the reference year, and assuming 52 paid weeks, the median workweek is approximately 37.5 hours, aligning with standard working hours in Canada. The median annual wage of $57,500 is higher than national estimates, although it has considerable variability. Immigrant respondents account for 19% of the observations, which is lower than national figures, potentially underrepresenting immigrant populations in the sample. Finally, both the maximum and minimum hourly wages fall within reasonable ranges, reinforcing the validity of the wage data.

Figure 1 shows a heavy right skew in the distribution of hourly wages, with the most common values in

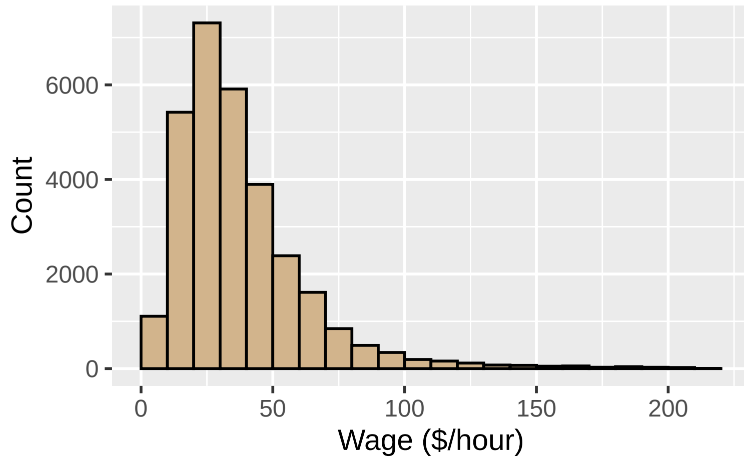

Figure 1: Distribution of Hourly Wages

the $20–40/hour wage range. There are around 1,000 observations in the $0–10/hour range, which is small but worth considering. Frequency decreases as hourly wages increase, and the distribution's shape makes estimating effects for high-wage earners more challenging.

4 Methodology

The selected model is Ordinary Least Squares with a log transformation applied. This transformation corrects the violation of the OLS assumption that the response variable (`hourly_wage`) is not normally distributed. Additionally, the log transformation offers a more convenient way to measure differences, as the estimates provide a way to represent the fixed effects as percentages rather than dollar values. The following equation represents the model used:

$$\ln(\text{hourly\_wage}) = \beta_0 + \beta_1 X + \beta_2(\text{imm\_length\_level}) + \beta_3(\text{imm}) + \epsilon,$$

where:

- $\ln($`hourly_wage`$)$ = The natural logarithm of `hourly_wage`.
- β_0 = The intercept term, representing the baseline expected log-hourly wage when all predictors are at their reference levels.
- X = A vector of control variables. (Listed in Table 2)
- β_1 = The vector of coefficients corresponding to the control variables in the vector X.

- `imm_length_level` = A categorical variable indicating how long the individual has been an immigrant.
- β_2 = The coefficient for the independent variable `imm_length_level`.
- `imm` = A binary variable indicating whether the individual is an immigrant.
- β_3 = The coefficient for the independent variable `imm`.
- ϵ = The error term.

5 Results

Table 2: Model Summary

	Dependent variable:		
	ln(Hourly Wage)		
	(1)	(2)	(3)
Constant	18.175^{***} (0.015)	14.732^{***} (0.014)	14.796^{***} (0.014)
pop_level	0.040^{***} (0.003)	0.017^{***} (0.003)	0.016^{***} (0.003)
age_level	0.059^{***} (0.001)	0.054^{***} (0.001)	0.052^{***} (0.001)
female	-0.109^{***} (0.007)	-0.157^{***} (0.006)	-0.158^{***} (0.006)
full_time	0.241^{***} (0.010)	0.096^{***} (0.010)	0.096^{***} (0.010)
highest_ed_level		0.237^{***} (0.004)	0.241^{***} (0.004)
imm_length_level			0.050^{***} (0.006)
imm	-0.038^{***} (0.009)	-0.098^{***} (0.009)	-0.194^{***} (0.016)
Province Controls	Yes	Yes	Yes
R^2	0.105	0.195	0.197
Residual Std. Error	0.582 (df = 30157)	0.552 (df = 30156)	0.552 (df = 30155)
Note:			*p<0.1; **p<0.05; ***p<0.01

Table 2 presents the summary of three regression models, with the intercept and coefficients presented on the natural scale to reflect baseline hourly wages and percentage changes, respectively. Model 1 includes all control variables except for highest education level and immigration length level. Model 2 adds the highest education level control but omits immigration length level. Model 3 includes all variables and controls as

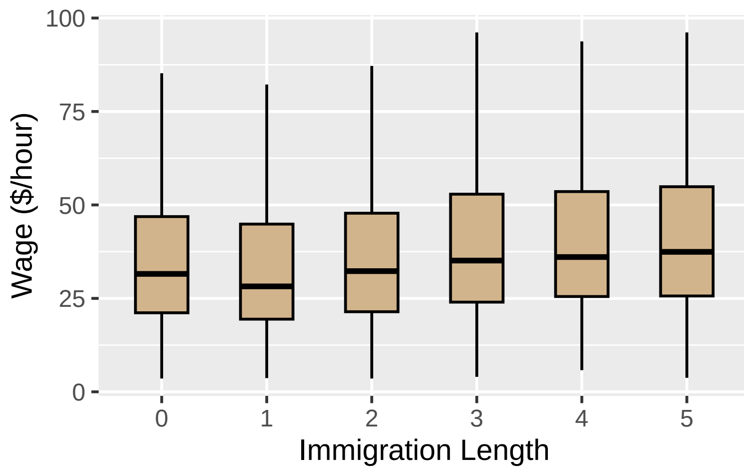

Figure 2: Hourly Wages by Immigration Length

outlined in Section 4.

Figure 2 shows the raw wage gap by immigration length. Initially, there is an observable wage difference between immigrants who have been in Canada for less than a decade and citizens. Wages tend to level off and even surpass those of citizens as immigrants become more integrated. Although the upward trend is in line with the theory and expectations, this would imply the raw gap is nearly zero, or slightly in favor of immigrants. However, this is not the case as Table 2 shows that the raw gap does not account for other control variables that must be considered.

In Model 1, without controlling for highest education level, non-native individuals earn approximately 3.8% less than natives on average. Although modest, this difference is statistically significant and supports the presence of a wage gap noted in previous studies. However, when compared to Model 3, where all controls are included, this gap increases by about 16 percentage points. This suggests omitted variable bias in Model 1. Since highest education level is positively correlated with both immigrant status and wages—particularly due to high-skilled immigration programs—the omission leads to a positive bias, causing the wage gap to be underestimated. When comparing Model 2 and Model 3, the addition of immigration length level does not substantially change the overall model fit. However, the coefficient on immigrant status still shifts by about 10 percentage points. This difference is not due to omitted variable bias in the traditional sense, as immigration length level is defined only for immigrants. Instead, its inclusion allows for contrasts between recent and long-term immigrants, changing the interpretation of the immigration status coefficient rather than correcting a bias.

Model 3 includes controls for province, age level, sex, and other relevant variables. In this model, the coefficient on `imm` is approximately -19.4%, indicating that, all else equal, immigrants earn about 19.4% less than native-born Canadians. This result is statistically and economically significant and closely matches the estimate reported in Lamb, Banerjee, and Verma (2021). The wage gap suggests immigrants are underpaid. While workplace discrimination is a possible factor, the literature more commonly attributes this to foreign credential recognition and language barriers, which are not included in the model and may contribute to the unexplained variation. The coefficient on `imm_length_level` is positive, around 5%, suggesting that immigrant wages increase over time, likely due to growing Canadian work experience. With `imm_length_level` at 4 (representing 30 to 39 years since arrival), the wage gap narrows to about 0.6%. At this stage, immigrants reach parity with natives and even have a slight wage advantage.

6 Discussion

The persistence of a wage gap between immigrants and native-born Canadians suggests that integration barriers are structural rather than temporary. Although the gap narrows over time, reaching near parity after 30–39 years, the delayed convergence limits the economic potential of newcomers in their most productive years. This lag points to shortcomings in credential recognition. Credential evaluation services like WES already bridge this gap with their equivalency statements, which translate foreign degrees into Canadian terms (Toronto Region Immigrant Employment Council 2025). However, more needs to be done to ensure employers actually understand and trust these evaluations. A strong policy move would be to create national training programs for HR professionals focused on interpreting international credentials. This would help make sure the human capital immigrants bring is properly valued, helping to lower skilled immigrant unemployment and easing labour shortages across Canada.

The results above have a few limitations related to both data and methodology. Although the data is from the 2022 tax year, it may not reflect changes in the labour market or immigration policy that occurred in 2023. Additionally, the dataset lacks key variables such as sector of employment, race, and language proficiency, which are important for capturing wage differences more accurately. Some variables—like years since immigration or population size—are categorized despite being naturally continuous, reducing analytical precision. Methodologically, the use of Ordinary Least Squares is prone to attenuation bias, which may lead to underestimation of the wage gap, as seen in Model 3 of Table 2. The inclusion of approximately seven control variables may also contribute to smaller observed wage gaps by overcontrolling the model. These factors should be considered when interpreting the results and their implications.

A potential avenue for further analysis could involve using inflation-adjusted GDP per capita for the time and region from which an individual emigrated. This variable is relevant as it provides valuable insight into the standard of living at that particular time, which is likely a strong determinant for an individual's decision to emigrate. As such, it meets the first requirement for a valid instrument: relevance. Additionally, the instrument does not directly affect wages except through its impact on the individual's immigration status. This meets the second requirement of a valid instrument: exogeneity. Given Canada's stringent labor regulations, which reduce the likelihood of discrimination based on immigrant status, this variable would not introduce direct bias into the analysis. However, public-use microdata, like the data used in this study, often lacks this type of detailed information. To ensure robustness, it is important that future models use more comprehensive sources, such as the Canadian Income Survey and Labour Force Survey master files, as utilized in Lamb, Banerjee, and Verma (2021).

7 Conclusion

In this paper, I find that immigrants in Canada earn about 19.4% less than native-born Canadians on average, but this gap shrinks steadily with time spent in the country, nearly disappearing after 30–39 years. This result matters because it shows that economic integration is a long-term process, and that initial barriers can slowly be overcome, even if they limit immigrants' earnings in their most productive years. Canada's immigration model has succeeded in many ways by attracting skilled individuals, but the slow wage convergence highlights the need for stronger policies that recognize foreign skills and education sooner. Supporting faster economic integration will not only make the system fairer but also strengthen Canada's overall growth and competitiveness.

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